

FOTIOS ILIOPOULOS

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CITIZENSHIP

Greek. Permanent Resident in the USA (Green Card holder).

CURRENT POSITION

Senior Research Scientist, Google May 2024 - present

PREVIOUS POSITIONS

Research Scientist, Google August 2021 - May 2024

Postdoctoral Scholar, Institute for Advanced Study and Princeton University Sep. 2019 - August 2021

EDUCATION

University of California Berkeley, USA Sep. 2014 - August 2019
Doctor of Philosophy (PhD) in Computer Science
Advisor: Professor Alistair Sinclair

National Technical University of Athens (NTUA), Greece Sep. 2008 - Sept. 2013
Diploma (5y) in Electrical and Computer Engineering

RESEARCH INTERESTS

Large Language Models, Efficient Deep Learning, Theoretical Computer Science.

PUBLICATIONS

1. “Gemma 4 Technical Report”, *technical report*, July 2026.
2. “Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities”, *technical report*, July 2025.
3. “Linear Projections of Teacher Embeddings for Few-Class Distillation” (with N. Loo, W. Hu, E. Vee), *technical report*, September. 2024.
4. “SLaM: Student-Label Mixing for Distillation with Unlabeled Examples” (with V. Kontonis, C. Baykal, K. Trihn, G. Menghani and E. Vee). *Proceedings of the 37th Conference on Neural Information Processing (NeurIPS)*, Nov. 2023.
5. “Robust Active Distillation (with C. Baykal, K. Trihn, G. Menghani and E. Vee).” *Proceedings of the 11th Conference on Learning Representations (ICLR)*, May 2023.
6. “Weighted distillation with unlabeled examples”. *Proceedings of the 36th Conference on Neural Information Processing (NeurIPS)*, December 2022, pp. 39:1-39:16.
7. “Improved bounds for coloring locally sparse hypergraphs”. *Proceedings of Approximation, Randomization, and Combinatorial Optimization (APPROX/RANDOM)*, August 2021, pp. 39:1-39:16.
8. “A new notion of commutativity for the algorithmic Lovász Local Lemma” (with D. G. Harris and V. Kolmogorov). *Theory of Computing (TOC)*, Volume 21, article 5. A preliminary version appeared in the proceedings of *Approximation, Randomization, and Combinatorial Optimization (APPROX/RANDOM)*, August 2021, pp. 31:1-31:25
9. “Group testing and local search: Is there a computational-statistical gap?” (with I. Zadik). *Proceedings of the 34th Annual Conference on Learning Theory (COLT)*, August 2021, pp. 2499-2551.
10. “Simple local computation algorithms for the Lovász Local Lemma” (with D. Achlioptas and T. Gouleakis). *Proceedings of the 32nd ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, July 2020, pp. 1-10.

11. “Efficiently list-edge coloring multigraphs asymptotically optimally” (with A. Sinclair). *Random Structures & Algorithms, Volume 61*, pp. 724-753, 2022. A preliminary version appeared in the proceedings of the 31st ACM-SIAM Symposium on Discrete Algorithms (SODA), January 2020, pp. 2319-2336.
12. “Beyond the Lovász Local Lemma: Point to set correlations and their algorithmic applications”, (with D. Achlioptas and A. Sinclair). *Proceedings of the 60th IEEE Symposium on Foundations of Computer Science (FOCS)*, November 2019, pp. 725-744.
13. “A local lemma for focused stochastic algorithms” (with D. Achlioptas and V. Kolmogorov). *SIAM Journal on Computing (SICOMP)*, Volume 48(5):1583-602; 2019.
14. “Commutative algorithms approximate the LLL-distribution”. *Proceedings of Approximation, Randomization, and Combinatorial Optimization (APPROX/RANDOM)*, August 2018, pp. 44:1 - 44:20.
15. “Stochastic control via entropy compression” (with D. Achlioptas and N. Vlassis). *Proceedings of the 44th International Colloquium on Automata, Language and Programming (ICALP)*, July 2017, pp. 83:1 - 83:13.
16. “Focused stochastic local search and the Lovász Local Lemma” (with D. Achlioptas). *Proceedings of the 27th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, January 2016, pp. 2024-2038.
17. “Random Walks that Find Perfect Objects and the Lovász Local Lemma” (with D. Achlioptas). *Journal of the ACM (J. ACM)*, Volume 63(3): 22:1-22:29; 2016. A preliminary version appeared in the proceedings of the 55th IEEE Annual Symposium on Foundations of Computer Science (FOCS), November 2014, pp. 494-503.

AWARDS

Christos Papakyriakopoulos award

For excellence in Mathematics in the school of Electrical and Computer Engineering for 2008-2009 (GPA 10/10)

December 2014

Nikos Kritikos Award

For excellence in Mathematics in the school of Electrical and Computer Engineering for 2008-2010 (GPA 10/10)

December 2011

Bronze Medal

Southeastern European Mathematical Olympiad for University Students
Member of the Greek National Team

March 2010

Silver Medal

Greek National Mathematical Olympiad

February 2008

3rd Prize

68 Panhellenic Contest in Mathematics

January 2008

PROGRAMMING

- Programming: Tensorflow, Pytorch, JAX, Python, C++, C, SQL, Matlab.

TEACHING EXPERIENCE

Teaching Assistant

- CS170: *Efficient Algorithms and Intractable Problems*, UC Berkeley (Spring 2019)
- CS70: *Discrete Mathematics and Probability Theory*, UC Berkeley (Spring 2018)
- CS174: *Combinatorics and Discrete Probability*, UC Berkeley (Spring 2017)
- *Discrete Mathematics for Computer Science*, National Technical University of Athens (Spring 2011)